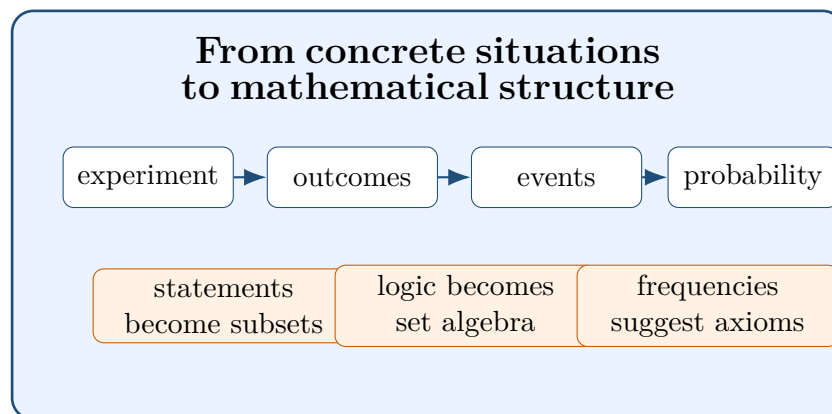


Discovering Formalism in Probability

General Lecture Notes



Prepared as a conceptual supplement to a task list on probability formalism

May 22, 2026

Contents

Preface	2
1 Why Formalism Is Needed	2
2 Random Experiments and Elementary Outcomes	2
2.1 Random experiments	3
2.2 Elementary outcomes	3
2.3 Sample spaces	3
3 Representation Is Not the Same as the Object	4
3.1 Lists	4
3.2 Tables and grids	4
3.3 Marked diagrams	4
4 Events as Subsets	5
4.1 Statements and events	5
4.2 Elementary outcomes versus events	5
4.3 The impossible and certain events	5
5 Logical Operations Become Set Operations	5
5.1 Union and intersection	6
5.2 Complement and difference	6
5.3 Disjoint events	6
5.4 Partitions	7
6 Set Identities and Their Meaning	7
6.1 Basic identities	7
6.2 De Morgan's laws	7
7 Graphical Thinking About Events	7
7.1 Why diagrams matter	8
7.2 Marking events in a grid	8
7.3 Reading a diagram as a statement	8
8 From Counts to Frequencies	9
8.1 Counting occurrences	9
8.2 Properties of observed frequencies	9
8.3 Why additivity requires disjointness	9
9 From Frequency to Probability	10
9.1 The abstraction step	10
9.2 Equally likely outcomes	10
9.3 Non-equally likely finite models	10
10 The Axiomatic Point of View	11
10.1 The general structure	11
10.2 Kolmogorov axioms	11
10.3 What comes from finite intuition?	12

11 Deriving Useful Probability Rules	12
11.1 Complement rule	12
11.2 Difference rule	12
11.3 Inclusion-exclusion	12
12 The Three Levels: Outcome, Data, Model	13
12.1 Level 1: elementary outcomes and events	13
12.2 Level 2: observed frequencies	13
12.3 Level 3: probability as an abstract model	13
13 Writing Correct Mathematical Explanations	13
13.1 A good explanation has structure	13
13.2 Examples of weak and strong reasoning	14
13.3 Common errors	14
14 Conceptual Examples	14
14.1 Example 1: two-stage quality status	14
14.2 Example 2: interpreting an overlap	15
14.3 Example 3: from observations to an axiom	15
15 Summary	15

Preface

These notes are not a collection of solved exercises. They are a structured lecture about the ideas that make elementary probability precise. The aim is to understand how one moves from a concrete situation, such as an observed process or a recorded experiment, to mathematical objects such as a sample space, an event, a set operation, a frequency, and finally a probability measure.

The central message is that probability theory is not only a set of formulas. It is a language for describing uncertain situations. The language becomes useful only when we know what the objects in the language mean.

Main theme

A probability model is built by a sequence of conceptual decisions:

experiment $\longrightarrow$ elementary outcomes $\longrightarrow$ sample space $\longrightarrow$ events $\longrightarrow$ probability.

Every later calculation depends on these earlier decisions.

1 Why Formalism Is Needed

In informal language we often say things such as:

- “the process succeeds”,
- “the result is acceptable”,
- “at least one alarm appears”,
- “the two observations agree”,
- “the outcome is not in the normal range”.

These statements are meaningful, but they are not yet mathematical objects. To calculate with them, compare them, combine them, or assign probabilities to them, we need to translate them into a precise form.

Probability formalism does exactly this. It replaces vague descriptions by sets and functions:

- a possible complete result becomes an *elementary outcome*;
- the collection of all possible elementary outcomes becomes a *sample space*;
- a statement about the result becomes an *event*, that is, a subset of the sample space;
- logical operations on statements become set operations on events;
- repeated observations lead to empirical frequencies;
- idealized long-run behavior is represented by probability.

Common misconception

Formalism is not decoration. It is not introduced to make simple ideas look complicated. It is introduced because without precise objects it is easy to count the wrong things, combine statements incorrectly, or confuse a concrete observation with an abstract model.

2 Random Experiments and Elementary Outcomes

2.1 Random experiments

Random experiment

A **random experiment** is a procedure whose possible results are known in advance, but whose actual result is not known before the experiment is performed.

The word “random” does not mean that the situation has no structure. It means that, at the time of modeling, the final result is uncertain. A random experiment may be highly structured. For example, a manufacturing inspection, a network request, a medical screening, or a sequence of choices in an application may all be modeled as random experiments.

2.2 Elementary outcomes

Elementary outcome

An **elementary outcome** is one complete possible result of the experiment, described at the level of detail chosen by the modeler.

The phrase “chosen by the modeler” is important. The same real situation can be modeled at different levels of detail.

Different levels of description

Suppose a device is inspected and the technician records its state.

- A coarse model may record only whether the device is *working* or *not working*. The elementary outcomes are then

$$\Omega = \{W, N\}.$$

- A more detailed model may record the state of two components: processor and battery. An elementary outcome may then be a pair such as

(processor OK, battery fail).

- An even more detailed model may record voltage, temperature, error code, and inspection time.

There is no unique sample space before the modeling convention is fixed.

2.3 Sample spaces

Sample space

The **sample space** Ω is the set of all elementary outcomes considered possible in the chosen model.

A sample space can be finite, countably infinite, or uncountable.

Type of sample space	Typical form	Example of outcome
Finite	$\Omega = \{\omega_1, \dots, \omega_n\}$	status from a finite list
Countably infinite	$\Omega = \{1, 2, 3, \dots\}$	waiting time measured in number of trials
Uncountable	$\Omega \subseteq \mathbb{R}$ or $\mathbb{R}^d$	measured temperature, time, position

This lecture focuses mainly on finite sample spaces, because they provide the clearest path from intuitive reasoning to probability axioms.

3 Representation Is Not the Same as the Object

A sample space can be represented by a list, a tree, a table, a coordinate system, a diagram, or a database table. These are representations of mathematical objects. They help us see structure, but they are not themselves the structure.

3.1 Lists

A list is useful when the number of elementary outcomes is small:

$$\Omega = \{a, b, c, d\}.$$

Lists are direct, but they often hide relations between outcomes. For example, if outcomes are pairs, a two-dimensional table may show the structure better than a linear list.

3.2 Tables and grids

A table is useful when an outcome has two coordinates. Suppose a system is checked twice, first in the morning and then in the evening. Each check can produce status N for normal or A for alarm. Then an elementary outcome has the form

$$\omega = (\text{morning status}, \text{evening status}).$$

A table can represent the sample space:

	evening N	evening A
morning N	(N, N)	(N, A)
morning A	(A, N)	(A, A)

This is not merely a convenient drawing. It shows that there are two coordinates and that each cell corresponds to one elementary outcome.

3.3 Marked diagrams

A marked diagram represents an event by marking those elementary outcomes for which a statement is true. For example, consider the event

$$E = \text{“at least one alarm is observed”}.$$

In the two-check model above, this event is marked as follows:

	evening N	evening A
morning N	.	X
morning A	X	X

The mathematical event is the set

$$E = \{(N, A), (A, N), (A, A)\}.$$

The table is a visual representation of the same subset.

Representation principle

A diagram is useful when it makes membership visible. A good representation should make it easy to answer the question: which elementary outcomes belong to the event?

4 Events as Subsets

4.1 Statements and events

Event

An **event** is a subset of the sample space. If Ω is the sample space, then an event is any set

$$A \subseteq \Omega.$$

The event occurs when the actual elementary outcome belongs to A .

A statement about an experiment becomes an event by collecting all elementary outcomes for which the statement is true.

From a statement to a set

Suppose

$$\Omega = \{L, M, H\} \times \{0, 1\},$$

where the first coordinate is risk level: low, medium, high; and the second coordinate says whether an alert appeared: 0 for no alert, 1 for alert.

The statement

“the risk is high or an alert appeared”

corresponds to the event

$$A = \{(H, 0), (H, 1), (L, 1), (M, 1)\}.$$

4.2 Elementary outcomes versus events

An elementary outcome is a single complete result. An event is a set of elementary outcomes. In a finite sample space, a one-element event has the form $\{\omega\}$, while the elementary outcome itself is ω .

Notation warning

Do not confuse ω with $\{\omega\}$. The first is an elementary outcome. The second is the event that this particular elementary outcome occurs.

4.3 The impossible and certain events

Two special events appear in every sample space:

- the **impossible event**, denoted $\emptyset$, contains no outcomes;
- the **certain event**, denoted Ω , contains all outcomes.

If the experiment is performed, exactly one elementary outcome occurs. Therefore the event Ω always occurs, while the event $\emptyset$ never occurs.

5 Logical Operations Become Set Operations

Statements can be combined using words such as “and”, “or”, “not”, and “but”. After translation into events, these become set operations.

Logical phrase	Set operation	Meaning
A and B	$A \cap B$	outcomes for which both statements are true
A or B	$A \cup B$	outcomes for which at least one statement is true
not A	A^c or $\Omega \setminus A$	outcomes for which A is false
A but not B	$A \setminus B = A \cap B^c$	outcomes in A and outside B
not both A and B	$(A \cap B)^c$	outcomes where at least one of the two fails
not A or B	$(A \cup B)^c$	outcomes outside both A and B

5.1 Union and intersection

Union and intersection

For events $A, B \subseteq \Omega$:

$$A \cup B = \{\omega \in \Omega : \omega \in A \text{ or } \omega \in B\},$$

$$A \cap B = \{\omega \in \Omega : \omega \in A \text{ and } \omega \in B\}.$$

The word “or” in mathematics usually means **inclusive or**: A or B or both.

Inclusive or

Let A mean “the request is urgent” and B mean “the request is expensive”. The event $A \cup B$ contains requests that are urgent, expensive, or both urgent and expensive. The overlap is not excluded.

5.2 Complement and difference

Complement and difference

The complement of A is

$$A^c = \Omega \setminus A = \{\omega \in \Omega : \omega \notin A\}.$$

The difference of A and B is

$$A \setminus B = \{\omega \in \Omega : \omega \in A \text{ and } \omega \notin B\}.$$

The phrase “ A but not B ” is one of the most important translations in elementary probability:

$$A \text{ but not } B \iff A \setminus B = A \cap B^c.$$

5.3 Disjoint events

Disjoint events

Events A and B are **disjoint** if they cannot occur together:

$$A \cap B = \emptyset.$$

Disjointness is a structural property of two events. It does not mean that one event has probability zero. It means that there is no elementary outcome that belongs to both events.

5.4 Partitions

Partition

A collection of events $A_1, \dots, A_k$ is a **partition** of Ω if:

1. every A_i is nonempty,
2. the events are pairwise disjoint: $A_i \cap A_j = \emptyset$ for $i \neq j$,
3. together they cover the whole sample space:

$$A_1 \cup \dots \cup A_k = \Omega.$$

A partition divides the sample space into non-overlapping cases. Many probability arguments are based on splitting a problem into such cases.

6 Set Identities and Their Meaning

Set identities are not abstract tricks. They express precise logical equivalences between statements.

6.1 Basic identities

For events $A, B, C \subseteq \Omega$:

$$\begin{aligned} A \cup B &= B \cup A, & A \cap B &= B \cap A, \\ (A \cup B) \cup C &= A \cup (B \cup C), & (A \cap B) \cap C &= A \cap (B \cap C), \\ A \cap (B \cup C) &= (A \cap B) \cup (A \cap C), \\ A \cup (B \cap C) &= (A \cup B) \cap (A \cup C), \\ A \cup A^c &= \Omega, & A \cap A^c &= \emptyset. \end{aligned}$$

6.2 De Morgan's laws

De Morgan's laws

For events A and B :

$$\begin{aligned} (A \cup B)^c &= A^c \cap B^c, \\ (A \cap B)^c &= A^c \cup B^c. \end{aligned}$$

In words: “not at least one” means “neither”, and “not both” means “at least one fails”.

A language example

Let A be “the customer chose express delivery” and B be “the customer used a discount code”.

- $(A \cup B)^c$ means: the customer did not choose express delivery and did not use a discount code.
- $(A \cap B)^c$ means: it is not true that the customer did both; perhaps neither happened, or exactly one happened.

These are different events.

7 Graphical Thinking About Events

7.1 Why diagrams matter

A diagram can make the structure of an event visible. This is especially important when outcomes have coordinates. A two-coordinate outcome naturally forms a grid. A multi-stage experiment naturally forms a tree. A continuous outcome may naturally form a region in the plane.

The purpose of a diagram is not decoration. It is to preserve the structure of the event and to prevent logical mistakes.

7.2 Marking events in a grid

Suppose two independent checks are performed on an item. Each check returns one of three labels:

$$G = \text{good}, \quad R = \text{requires review}, \quad F = \text{failed}.$$

An elementary outcome is a pair

(first check, second check).

Consider the statement:

$$E = \text{“at least one check failed”}.$$

The marked grid is:

	second G	second R	second F
first G	.	.	X
first R	.	.	X
first F	X	X	X

The event is

$$E = \{(G, F), (R, F), (F, G), (F, R), (F, F)\}.$$

How to mark a diagram

To mark an event, inspect each elementary outcome and ask: does the statement become true for this outcome? If yes, mark it. If no, leave it unmarked.

7.3 Reading a diagram as a statement

Going in the opposite direction is equally important. Suppose a diagram is given and we need to describe the event. The goal is not merely to list marked cells. The goal is to find the underlying statement.

Consider the marked grid:

	second G	second R	second F
first G	X	.	.
first R	.	X	.
first F	.	.	X

The marked cells are (G, G) , (R, R) , and (F, F) . The conceptual description is:

“the two checks give the same label.”

This statement is more informative than a bare list because it reveals the pattern.

8 From Counts to Frequencies

8.1 Counting occurrences

Suppose an experiment is repeated n times and the results are recorded. For an event A , let

$$n(A)$$

be the number of repetitions in which event A occurred.

Observed frequency

The **observed frequency** or **empirical relative frequency** of an event A is

$$f_n(A) = \frac{n(A)}{n},$$

where n is the total number of repetitions.

Observed frequency is data-based. It depends on the concrete repetitions that actually happened.

8.2 Properties of observed frequencies

For any finite set of observed trials, observed frequencies satisfy several important properties.

Frequency properties

For events $A, B \subseteq \Omega$:

1. $0 \leq f_n(A) \leq 1$.
2. $f_n(\emptyset) = 0$.
3. $f_n(\Omega) = 1$.
4. If $A \cap B = \emptyset$, then

$$f_n(A \cup B) = f_n(A) + f_n(B).$$

5. For every event A ,

$$f_n(A^c) = 1 - f_n(A).$$

These properties do not require any advanced probability theory. They follow from counting.

8.3 Why additivity requires disjointness

If two events overlap, then simply adding their frequencies counts the overlap twice. The correct relation is the empirical version of inclusion-exclusion:

$$f_n(A \cup B) = f_n(A) + f_n(B) - f_n(A \cap B).$$

Overlapping categories

In a dataset of service requests, let

$$A = \text{“request is urgent”}, \quad B = \text{“request is from a premium client”}.$$

If some requests are both urgent and from premium clients, then A and B overlap. Adding $f_n(A)$ and $f_n(B)$ counts those requests twice. Subtracting $f_n(A \cap B)$ corrects the double counting.

9 From Frequency to Probability

Observed frequency is a property of data. Probability is a property of a mathematical model. The two are connected, but they are not the same object.

9.1 The abstraction step

When we observe frequencies, we notice stable algebraic patterns: non-negativity, normalization, additivity for disjoint events, complement rules, and inclusion-exclusion. Probability abstracts these patterns into a mathematical function.

Probability on a finite sample space

Let Ω be a finite sample space. A **probability measure** on Ω is a function

$$P : \mathcal{P}(\Omega) \rightarrow [0, 1]$$

such that:

1. $P(\Omega) = 1$,
2. if $A \cap B = \emptyset$, then

$$P(A \cup B) = P(A) + P(B).$$

Here $\mathcal{P}(\Omega)$ denotes the set of all subsets of Ω .

In a finite sample space, the probability of an event is determined by the probabilities of elementary outcomes:

$$P(A) = \sum_{\omega \in A} P(\{\omega\}).$$

The elementary probabilities must satisfy

$$P(\{\omega\}) \geq 0, \quad \sum_{\omega \in \Omega} P(\{\omega\}) = 1.$$

9.2 Equally likely outcomes

If all elementary outcomes in a finite sample space are equally likely, then

$$P(A) = \frac{|A|}{|\Omega|}.$$

This is the classical formula. It is powerful, but only under the assumption that all elementary outcomes are equally likely.

The classical formula is not universal

The formula $P(A) = |A|/|\Omega|$ is valid only when all elementary outcomes are equally likely. If outcomes have different probabilities, counting favorable outcomes is not enough.

9.3 Non-equally likely finite models

A finite model may assign different probabilities to different outcomes. For example, suppose a server response is classified as

$$\Omega = \{\text{fast}, \text{normal}, \text{slow}, \text{failed}\}.$$

A model could assign

$$P(\text{fast}) = 0.20, \quad P(\text{normal}) = 0.65, \quad P(\text{slow}) = 0.12, \quad P(\text{failed}) = 0.03.$$

Then for

$$A = \{\text{slow}, \text{failed}\},$$

we get

$$P(A) = 0.12 + 0.03 = 0.15.$$

The probability is found by adding assigned probabilities, not by dividing the number of favorable outcomes by the number of all outcomes.

10 The Axiomatic Point of View

10.1 The general structure

Modern probability theory is formulated using three ingredients:

$$(\Omega, \mathcal{F}, P).$$

Probability space

A **probability space** is a triple

$$(\Omega, \mathcal{F}, P),$$

where:

- Ω is the sample space,
- $\mathcal{F}$ is a collection of events, usually a σ -algebra of subsets of Ω ,
- P is a probability measure assigning probabilities to events in $\mathcal{F}$.

In finite probability, one usually takes

$$\mathcal{F} = \mathcal{P}(\Omega),$$

the set of all subsets of Ω . In more advanced probability, especially for uncountable spaces, not every subset must be treated as an event.

10.2 Kolmogorov axioms

Kolmogorov axioms

A probability measure P satisfies:

1. **Non-negativity:**

$$P(A) \geq 0 \quad \text{for every event } A.$$

2. **Normalization:**

$$P(\Omega) = 1.$$

3. **Countable additivity:** if $A_1, A_2, A_3, \dots$ are pairwise disjoint events, then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

For finite sample spaces, countable additivity reduces essentially to finite additivity. Its deeper meaning appears when infinitely many events are involved.

10.3 What comes from finite intuition?

Finite experiments strongly suggest:

- probabilities should not be negative;
- the whole sample space should have probability 1;
- disjoint events should have additive probabilities;
- complements should satisfy $P(A^c) = 1 - P(A)$.

These mirror the behavior of observed frequencies.

Countable additivity is more subtle. It is not directly forced by observing a finite number of repetitions, because finite data never requires us to add infinitely many disjoint parts. Countable additivity is a mathematical strengthening needed for a coherent theory of infinite processes and continuous models.

11 Deriving Useful Probability Rules

The axioms imply many familiar rules. These rules should not be memorized as isolated formulas; they follow from the structure of events.

11.1 Complement rule

Since A and A^c are disjoint and

$$A \cup A^c = \Omega,$$

we have

$$P(A) + P(A^c) = P(\Omega) = 1.$$

Thus

$$P(A^c) = 1 - P(A).$$

11.2 Difference rule

The event A can be split into two disjoint parts:

$$A = (A \cap B) \cup (A \cap B^c).$$

Therefore

$$P(A) = P(A \cap B) + P(A \cap B^c),$$

so

$$P(A \setminus B) = P(A \cap B^c) = P(A) - P(A \cap B).$$

11.3 Inclusion-exclusion

The union $A \cup B$ consists of three disjoint regions:

$$A \setminus B, \quad A \cap B, \quad B \setminus A.$$

Adding $P(A)$ and $P(B)$ counts $A \cap B$ twice. Hence

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

When simple addition is valid

The formula

$$P(A \cup B) = P(A) + P(B)$$

is valid only when A and B are disjoint. Otherwise the overlap must be subtracted.

12 The Three Levels: Outcome, Data, Model

A central skill in probability is to distinguish three levels of thinking.

12.1 Level 1: elementary outcomes and events

At the first level, we ask:

- What is one complete result of the experiment?
- What is the sample space?
- Which outcomes make a statement true?
- How do statements combine as sets?

This level is purely structural. It can be studied before assigning any numerical probabilities.

12.2 Level 2: observed frequencies

At the second level, we repeat or observe the experiment and count how often events occur. Frequencies are empirical. They summarize a concrete dataset.

Important facts about frequencies:

- they can change from one dataset to another;
- they satisfy algebraic rules because they are based on counts;
- they suggest the rules that probabilities should obey.

12.3 Level 3: probability as an abstract model

At the third level, we introduce a mathematical function P that assigns probabilities to events. This function is not a dataset. It is part of a model.

The relationship between levels is:

events define what is counted, frequencies describe data, probabilities describe a model.

Do not collapse the levels

It is incorrect to treat one observed frequency as automatically equal to the true probability. A frequency is a numerical summary of observed data. A probability is an idealized model value.

13 Writing Correct Mathematical Explanations

This part is especially important in conceptual probability. A correct final number is not enough if the reasoning is unclear.

13.1 A good explanation has structure

A complete explanation usually contains:

1. a clear description of the experiment;
2. the sample space or at least the form of elementary outcomes;
3. the event expressed as a subset or as a clear condition on outcomes;
4. the relevant set operation or probability rule;
5. a justification for why the rule applies;
6. a final interpretation in words.

13.2 Examples of weak and strong reasoning

Weak reasoning

“Add the two probabilities because there are two conditions.”

This is weak because it does not check whether the events are disjoint. Addition is not automatically valid.

Stronger reasoning

“The two events are disjoint because no elementary outcome can satisfy both conditions at the same time. Therefore the probability of their union is the sum of their probabilities.”

This is stronger because it connects the formula to the structure of the events.

13.3 Common errors

Error	Why it is a problem
Listing outcomes without explaining the pattern	The structure of the event remains hidden.
Using $P(A \cup B) = P(A) + P(B)$ without checking disjointness	The overlap may be counted twice.
Confusing “or” with exclusive or	Mathematical union includes the case where both statements are true.
Confusing elementary outcomes with events	ω and $\{\omega\}$ are different objects.
Treating empirical frequency as exact probability	Data and model are different levels.
Using a formula before defining the sample space	The formula may count the wrong objects.

14 Conceptual Examples

The following examples are included to illustrate the general structures. They are not solutions to any particular task list problem.

14.1 Example 1: two-stage quality status

A product is tested by two automatic sensors. Each sensor returns one of two statuses:

$$P = \text{pass}, \quad F = \text{fail}.$$

The elementary outcomes are pairs:

$$\Omega = \{(P, P), (P, F), (F, P), (F, F)\}.$$

Let

$$A = \text{“the first sensor fails”},$$

$B = \text{“at least one sensor fails”}.$

Then

$$A = \{(F, P), (F, F)\},$$

$$B = \{(P, F), (F, P), (F, F)\}.$$

We have $A \subseteq B$, because whenever the first sensor fails, at least one sensor fails.

14.2 Example 2: interpreting an overlap

Let

$A = \text{“an order is international”},$

$B = \text{“an order is delayed”}.$

Then:

- $A \cap B$ means: international and delayed;
- $A \setminus B$ means: international but not delayed;
- $A^c \cap B$ means: not international but delayed;
- $A^c \cap B^c$ means: neither international nor delayed.

The four regions form a partition of the sample space.

14.3 Example 3: from observations to an axiom

Suppose 500 requests were observed. Let A be the event “request was handled by chat” and B the event “request was handled by phone”. If each request is handled by exactly one channel, then A and B are disjoint. If 180 requests were handled by chat and 120 by phone, then

$$f_{500}(A) = \frac{180}{500}, \quad f_{500}(B) = \frac{120}{500},$$

$$f_{500}(A \cup B) = \frac{300}{500} = \frac{180}{500} + \frac{120}{500}.$$

This is not merely arithmetic. It shows why additivity for disjoint events is natural.

15 Summary

The movement from intuition to formalism can be summarized as follows.

1. A random experiment must be described at a chosen level of detail.
2. An elementary outcome is one complete possible result at that level of detail.
3. The sample space Ω is the set of all elementary outcomes.
4. An event is a subset of Ω .
5. Statements about outcomes become events by collecting all outcomes that make the statement true.
6. Logical operations become set operations: “and” becomes intersection, “or” becomes union, and “not” becomes complement.
7. Graphical representations are valuable when they reveal the structure of membership, union, intersection, complement, and difference.
8. Observed frequencies are based on counts from data and naturally satisfy non-negativity, normalization, complement rules, and additivity for disjoint events.
9. Probability abstracts these frequency-like properties into a mathematical measure.

10. The axiomatic approach clarifies which rules are assumptions of the model and which are consequences of those assumptions.

Final principle

To understand probability, first understand the set structure. Probability calculations are reliable only when the events being calculated have been correctly identified.